

PlantPulse — Incident Report

PlantPulse Demo Site — Process Unit 7 · Generated 2026-07-22 18:52

Incident	Early Bearing Failure on an Overdue-Maintenance Compressor
Zone	Z2
Window	2026-07-22T09:00:00 → 2026-07-22T13:00:00
Ground-truth severity	HIGH
Single-channel alarm would fire?	No — this is exactly the compound pattern PlantPulse exists to catch

Compound Risk at Peak

83 / 100

Confidence: 47%

Contributing Factors

- Equipment: +85%
- Sensor: +12%
- Weather: +3%

Root Cause (Knowledge Graph)

COMP-01 shows Compressor Temperature rising, Vibration rising; MT-01 (Bearing inspection & lubrication) is overdue.

SOP Response — SOP-MECH-01: Rotating Equipment Vibration / Bearing Alarm Response

Source: OISD process safety management framework — mechanical integrity / predictive maintenance practice for rotating equipment

- 1 Cross-check the vibration and bearing-temperature trend against baseline to confirm this is a genuine trend, not a sensor fault
- 2 Reduce load or bring the equipment to a controlled shutdown if the trend continues to climb
- 3 Escalate the overdue maintenance task to the planner for priority scheduling — do not defer it further
- 4 Perform a manual bearing and lubrication inspection before restart
- 5 Update the asset's maintenance record and shorten the inspection interval going forward

Estimated risk reduction if followed: 77% (peak compound risk 83 → 19, computed by re-scoring a mitigated counterfactual through the same fusion model)

Detection Performance

Across 20 independently seeded evaluation days, PlantPulse detected this archetype in **100%** of instances with a median lead time of **100 minutes** over an extrapolated conventional single-sensor alarm. Source: engine/metrics/results.json.

Generated by PlantPulse from simulated scenario data (seed-reproducible). This is a demonstration report for the ET AI Hackathon 2.0 submission, not a record of a real industrial incident.